

BOLZANO, ItalyWMO: **160200**Lat: **46.460N**Lon: **11.326E**Elev: **241**StdP: **98.47**Time Zone: **1.00 (EUC)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-7.2	-5.9	-17.6	0.8	2.0	-14.4	1.1	2.6	7.7	5.9	6.3	7.1	0.8	150	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	33.8	21.4	32.1	20.8	30.7	20.3	22.6	31.3	21.8	30.1	21.1	28.8	3.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	15.0	25.6	19.1	14.3	24.6	18.3	13.6	23.9	67.3	31.2	64.5	30.4	62.0	28.9	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.4	5.4	DB -9.9	35.8	1.1	2.4	-10.7	37.5	-11.4	38.9	-12.0	40.2	-12.8	41.9		
(5)			WB -10.4	23.8	1.3	0.6	-11.3	24.3	-12.1	24.6	-12.8	25.0	-13.7	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	1.7	4.3	9.7	13.8	17.7	22.1	23.4	22.9	18.4	13.3	6.8	1.6
	DBStd	8.36	2.65	2.92	3.34	3.28	3.02	3.25	2.69	3.01	2.83	3.40	3.42	3.08
	HDD10.0	844	257	159	46	6	0	0	0	0	0	10	106	260
	HDD18.3	2429	515	392	267	139	48	7	2	3	33	158	347	518
	CDD10.0	1951	0	1	36	120	240	363	416	400	252	113	9	0
	CDD18.3	495	0	0	0	3	30	121	159	144	35	3	0	0
	CDH23.3	4688	0	0	7	72	316	1173	1542	1318	250	11	0	0
	CDH26.7	1776	0	0	0	10	79	468	636	536	47	0	0	0
Wind	WSAvg	1.6	1.0	1.4	1.9	2.1	2.2	2.3	2.1	1.9	1.5	1.1	1.0	1.0
Precipitation	PrecAvg	725	27	27	47	49	84	73	93	100	73	63	56	28
	PrecMax	1074	122	92	115	86	192	144	239	254	258	198	168	88
	PrecMin	420	0	0	0	1	26	12	19	24	3	0	0	0
	PrecStd	160	26	26	33	25	40	32	48	56	68	58	45	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	17.5	24.1	28.0	31.1	35.0	35.3	36.3	30.4	24.8	17.8	12.2
		MCWB	6.6	8.5	12.0	15.3	19.0	22.0	21.8	21.8	20.3	17.1	13.4	6.0
	2%	DB	10.1	14.0	21.1	25.2	28.9	33.0	33.2	33.8	28.0	22.2	15.0	10.1
		MCWB	4.2	6.8	11.1	14.8	18.0	21.0	21.2	21.5	19.2	15.8	11.1	5.3
	5%	DB	8.1	12.0	19.0	23.1	26.8	31.2	32.0	31.2	26.0	20.9	13.2	8.1
		MCWB	3.5	5.7	10.1	13.6	17.1	20.6	20.6	20.7	17.8	15.4	9.8	4.7
	10%	DB	6.2	10.0	16.8	21.1	24.8	29.5	30.2	29.8	24.2	19.0	12.0	6.6
		MCWB	2.4	4.7	9.4	12.9	16.0	19.8	20.2	20.3	17.2	14.2	9.3	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	9.3	12.9	16.6	20.6	23.6	23.4	23.2	21.4	18.3	14.1	8.5
		MCDB	11.9	16.6	22.6	25.4	28.4	31.4	32.4	31.9	28.7	23.0	16.6	9.8
	2%	WB	5.6	7.6	11.7	15.5	19.0	22.3	22.3	22.4	20.1	17.0	12.4	6.6
		MCDB	8.3	11.7	19.2	23.7	26.9	30.9	30.9	31.2	25.6	20.8	14.0	8.1
	5%	WB	4.2	6.5	10.8	14.4	17.9	21.4	21.6	21.5	19.1	16.0	11.2	5.4
		MCDB	6.6	10.0	17.3	21.7	24.7	29.6	29.8	29.7	24.0	19.3	12.5	7.1
	10%	WB	3.2	5.5	9.9	13.4	16.9	20.5	21.0	20.8	18.2	15.0	10.1	4.2
		MCDB	5.4	8.6	15.4	19.5	23.0	28.0	28.7	28.0	22.8	17.9	11.3	6.2
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.8	12.5	12.2	12.3	12.8	12.9	12.6	11.5	9.9	9.1	9.2
		MCDBR	11.6	14.2	17.2	16.0	15.5	15.2	15.2	15.2	13.6	11.9	11.0	11.0
		MCWBR	7.2	8.0	8.7	7.3	6.5	5.7	5.4	5.3	6.1	6.2	7.0	7.1
	5% WB	MCDBR	10.5	11.8	14.7	14.5	13.6	14.0	13.5	13.5	11.7	10.1	8.1	9.2
		MCWBR	6.9	6.9	7.5	6.9	5.9	5.7	5.4	5.2	5.4	5.5	5.7	6.3
Clear-Sky Solar Irradiance	taub		0.297	0.332	0.393	0.457	0.472	0.488	0.478	0.468	0.431	0.398	0.342	0.300
	taud		2.325	2.234	2.146	2.053	2.074	2.094	2.136	2.178	2.247	2.290	2.390	2.399
	Ebn at Noon		791	826	820	798	800	788	791	781	775	742	730	748
	Edh at Noon		76	102	129	156	158	156	148	137	117	97	72	64
All-Sky Solar Radiation	RadAvg		1.64	2.59	3.85	4.54	4.95	5.66	5.73	4.99	3.87	2.58	1.58	1.36
	RadStd		0.18	0.29	0.47	0.63	0.52	0.41	0.41	0.44	0.37	0.34	0.26	0.17

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (85 neighbors)		+0.36	N/A	N/A	+0.63	+0.36	+0.31	N/A	N/A	+66	+16	

Nomenclature: See separate page